

OLSZTYN, Poland

WMO: 122720

Lat: 53.7686N

Lon: 20.4213E

Elev: 137

StdP: 99.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-13.3	-19.3	0.7	-16.0	-16.1	0.9	-12.6	9.2	0.0	8.3	0.1	1.8	120	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	28.9	19.6	27.0	18.7	25.2	17.9	21.1	26.6	20.0	24.9	19.0	23.4	3.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.2	23.2	18.2	13.3	22.3	17.2	12.5	21.2	61.5	26.6	57.9	25.0	54.5	23.5	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.7	6.0	-20.5	32.2	4.3	1.7	-23.7	33.4	-26.2	34.4	-28.7	35.4	-31.8	36.7		
(5)			-20.8	22.8	4.2	1.1	-23.9	23.6	-26.3	24.3	-28.7	24.9	-31.8	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-2.6	-1.4	2.1	7.8	12.6	15.8	18.3	17.6	13.2	8.2	3.4	-0.5
	DBStd	8.40	5.55	5.24	4.35	4.33	4.12	3.42	3.15	2.92	3.24	3.79	3.95	5.11
	HDD10.0	1683	389	319	247	92	19	1	0	0	9	81	199	327
	HDD18.3	3888	647	553	504	317	183	91	41	49	156	314	448	585
	CDD10.0	927	0	0	1	26	101	176	256	237	106	24	1	0
	CDD18.3	89	0	0	0	1	6	15	39	27	2	0	0	0
	CDH23.3	854	0	0	0	14	78	166	344	231	22	0	0	0
	CDH26.7	194	0	0	0	2	9	39	96	46	2	0	0	0
Wind	WSAvg	2.9	3.3	3.2	3.3	2.9	2.9	2.8	2.7	2.4	2.5	2.9	3.1	3.2
Precipitation	PrecAvg	657	44	35	41	39	59	69	88	68	58	56	53	47
	PrecMax	784	123	63	86	106	141	120	226	154	178	144	110	85
	PrecMin	451	6	5	8	9	24	26	13	13	19	7	13	12
	PrecStd	84	24	14	20	23	23	26	46	33	37	36	24	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	9.6	16.1	25.4	27.9	30.6	32.0	30.4	26.2	20.0	12.9	10.0
		MCWB	7.0	7.2	9.7	15.4	17.7	20.5	20.9	21.2	18.0	14.9	11.2	9.0
	2%	DB	6.7	8.1	12.9	21.4	25.6	27.4	29.3	27.8	23.2	17.5	11.0	8.2
		MCWB	5.7	6.3	8.1	13.3	16.8	18.8	20.2	19.4	16.7	13.7	9.5	7.0
	5%	DB	5.3	6.5	10.3	18.3	23.3	25.0	27.0	25.9	21.0	15.5	9.6	6.9
		MCWB	4.3	5.1	7.1	11.3	15.6	17.3	19.2	18.4	15.5	12.8	8.4	6.0
	10%	DB	3.9	5.1	8.3	15.7	20.8	23.0	24.9	24.1	19.1	13.8	8.4	5.5
		MCWB	3.1	3.8	5.7	10.0	14.3	16.5	18.5	17.7	14.7	11.7	7.4	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	8.0	10.7	15.9	19.0	21.7	22.5	22.1	18.9	16.0	11.6	9.1
		MCDB	8.3	9.0	14.7	24.3	25.5	27.9	28.7	27.7	24.0	19.4	12.5	9.8
	2%	WB	5.8	6.6	8.9	13.9	17.6	19.8	21.3	20.6	17.3	14.3	9.9	7.3
		MCDB	6.6	7.7	11.7	19.9	23.6	25.3	27.2	25.6	21.5	16.5	10.8	7.9
	5%	WB	4.4	5.3	7.3	12.1	16.4	18.5	20.3	19.5	16.2	13.1	8.5	6.0
		MCDB	5.2	6.5	9.8	17.0	22.2	23.4	25.3	24.0	19.7	15.1	9.4	6.8
	10%	WB	3.1	3.8	5.9	10.6	15.1	17.3	19.3	18.5	15.2	11.9	7.5	4.8
		MCDB	3.8	4.9	8.1	14.7	19.7	21.6	23.5	22.6	18.2	13.7	8.3	5.5
Mean Daily Temperature Range	5% DB	MDBR	4.3	5.3	7.1	10.1	10.8	10.0	9.7	9.9	8.6	6.7	4.1	3.9
		MCDBR	4.3	6.2	10.7	14.8	14.3	13.7	13.2	13.1	12.2	9.3	5.4	4.3
		MCWBR	3.9	4.9	6.8	7.7	6.8	6.4	5.6	5.9	6.4	6.0	4.4	3.9
	5% WB	MCDBR	4.3	5.7	9.6	13.2	12.8	11.9	11.5	11.0	10.4	8.4	5.0	4.3
		MCWBR	3.9	4.9	6.7	7.4	6.7	6.5	5.5	5.5	5.8	5.7	4.4	4.0
Clear-Sky Solar Irradiance	taub	0.302	0.332	0.365	0.410	0.399	0.398	0.430	0.420	0.377	0.357	0.343	0.301	
	taud	2.228	2.237	2.248	2.190	2.281	2.313	2.241	2.270	2.368	2.412	2.310	2.263	
	Ebn at Noon	645	749	806	814	848	852	813	798	790	722	593	571	
	Edh at Noon	63	86	106	128	124	121	128	118	94	73	60	51	
All-Sky Solar Radiation	RadAvg	0.61	1.32	2.55	4.13	5.29	5.51	5.09	4.38	3.11	1.63	0.62	0.41	
	RadStd	0.11	0.22	0.41	0.54	0.51	0.49	0.56	0.41	0.41	0.28	0.07	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-142	+115	N/A
(47) Regional (25 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	-146	+74	N/A

Nomenclature: See separate page